Introduction and learning objectives

In Lecture 11, we trained a diffusion model to predict added noise. In this tutorial, we derive two alternatives: predicting the clean image and predicting the score. We then compare their training losses.

Recall from Lecture 11. Let $\mathbf{x}_0$ be a clean image from the training data. At noise level $t \in \{1, \dots, T\}$, add independent Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\mathbf{z}_t = a_t \mathbf{x}_0 + b_t \boldsymbol{\epsilon}, \qquad q(\mathbf{z}_t \mid \mathbf{x}_0) = \mathcal{N}(a_t \mathbf{x}_0, b_t^2 \mathbf{I}),$$

where $a_t = \sqrt{\bar{\alpha}_t}$, $b_t = \sqrt{1 - \bar{\alpha}_t}$ and $\bar{\alpha}_t = \prod_{r=1}^{t} \alpha_r$, with $0 < \alpha_r < 1$. The network takes the noisy image $\mathbf{z}_t$ and noise level t as inputs.

Problem 1. Predicting the clean image.

(a) Rearrange the forward equation and replace the unknown noise by its prediction $\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)$ to obtain

$$\hat{\mathbf{x}}_\theta(\mathbf{z}_t, t) = \frac{\mathbf{z}_t - b_t \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)}{a_t}.$$

(b) For a fixed noisy image $\mathbf{z}_t$ and noise level t, choose a predicted image $\mathbf{u}$ to minimise the expected squared error

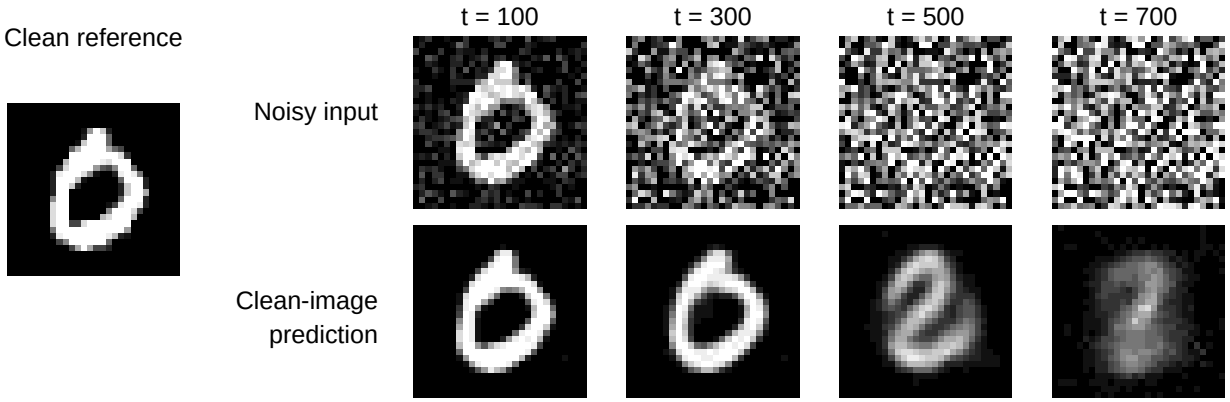
$$R(\mathbf{u}) = \mathbb{E}[\|\mathbf{x}_0 - \mathbf{u}\|^2 \mid \mathbf{z}_t, t].$$

Treat the pixel values in $\mathbf{u}$ as the variables to optimise. Show that the minimum occurs at

$$\mathbf{x}^*(\mathbf{z}_t, t) = \mathbb{E}[\mathbf{x}_0 \mid \mathbf{z}_t, t].$$

Steps: Expand the squared error, take the conditional expectation, differentiate with respect to $\mathbf{u}$, and set the gradient to zero.

(c) Describe the change from $t = 100$ to $t = 700$ below. Explain it using the loss $R(\mathbf{u})$ and prediction $\mathbf{x}^*(\mathbf{z}_t, t)$ from part (b). Is the prediction a sample from the conditional distribution of clean images?



Lecture 11 model outputs, converted using part (a). Each column uses the same clean image and noise, with a different noise level. Pixel values are clipped for display. Each prediction is made directly from its noisy input.

Problem 2. Predicting the score. The *score* is the gradient of a log density with respect to its input. Let $q_t(\mathbf{z}_t)$ be the density of noisy images at level t, and write

$$\mathbf{s}_t^*(\mathbf{z}_t) = \nabla_{\mathbf{z}_t} \log q_t(\mathbf{z}_t).$$

(a) First hold the clean image $\mathbf{x}_0$ fixed. Starting from the Gaussian log density, show that its *conditional score* is

$$\mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0) := \nabla_{\mathbf{z}_t} \log q(\mathbf{z}_t \mid \mathbf{x}_0) = -\frac{\mathbf{z}_t - a_t \mathbf{x}_0}{b_t^2} = -\frac{\boldsymbol{\epsilon}}{b_t}.$$

Steps: Write the log density as a squared-error term plus a constant. Differentiate with respect to $\mathbf{z}_t$, then substitute the forward equation. Which point does this vector point towards?

(b) We can train a network $\mathbf{s}_\theta(\mathbf{z}_t, t)$ to predict $\mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0)$ using

$$J_s(\theta) = \mathbb{E}_{\mathbf{x}_0, t, \boldsymbol{\epsilon}} \left[\left\| \mathbf{s}_\theta(\mathbf{z}_t, t) + \frac{\boldsymbol{\epsilon}}{b_t} \right\|^2 \right].$$

Use the result from Problem 1(b) to show that, for fixed $\mathbf{z}_t$ and t, the prediction that minimises the expected loss is

$$\mathbf{s}^{\text{opt}}(\mathbf{z}_t, t) = -\frac{\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{z}_t, t]}{b_t} = \mathbf{s}_t^*(\mathbf{z}_t).$$

For the last equality, you may use the identity

$$\mathbb{E}[\mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0) \mid \mathbf{z}_t, t] = \nabla_{\mathbf{z}_t} \log q_t(\mathbf{z}_t).$$

Optional A gives the proof. Explain how the training target $\mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0)$ differs from the score $\mathbf{s}_t^*(\mathbf{z}_t)$.

(c) A noise predictor also gives a score estimate: $\mathbf{s}_\theta(\mathbf{z}_t, t) = -\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)/b_t$. In Lecture 11, the reverse mean was

$$\boldsymbol{\mu}_\theta(\mathbf{z}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{z}_t - \frac{1 - \alpha_t}{b_t} \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t) \right).$$

Substitute the score estimate to obtain

$$\boxed{\boldsymbol{\mu}_\theta(\mathbf{z}_t, t) = \frac{1}{\sqrt{\alpha_t}} \big(\mathbf{z}_t + (1 - \alpha_t) \mathbf{s}_\theta(\mathbf{z}_t, t) \big).}$$

Do we need to train another network to use this form? For $t \geq 2$, the sampler uses

$$\mathbf{z}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{z}_t, t) + \sigma_t \boldsymbol{\xi}_t,$$

with new, independent noise $\boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and fixed reverse-step variance $\sigma_t^2 > 0$. Does converting the output change the distribution of $\mathbf{z}_{t-1}$?

Problem 3. Comparing the training losses. Compare predictions related by the formulas in Problems 1(a) and 2(c). Define the errors for one training example:

$$L_\epsilon = \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)\|^2,$$
$$L_x = \|\mathbf{x}_0 - \hat{\mathbf{x}}_\theta(\mathbf{z}_t, t)\|^2,$$
$$L_s = \|\mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0) - \mathbf{s}_\theta(\mathbf{z}_t, t)\|^2.$$

(a) Show that

$$\boxed{L_\epsilon = \frac{a_t^2}{b_t^2} L_x = b_t^2 L_s.}$$

Steps: Substitute $\mathbf{z}_t = a_t \mathbf{x}_0 + b_t \boldsymbol{\epsilon}$ into the clean-image prediction and subtract $\mathbf{x}_0$. Similarly, subtract the score target from the score prediction. Square the two error relations.

(b) The lecture's simple noise loss is $J_\epsilon(\theta) = \mathbb{E}[L_\epsilon]$, with t sampled uniformly. What weights should multiply L_x and L_s to give the same expected loss? If we instead use *unweighted* L_x or L_s, do all three expected losses still agree? Which noise levels get larger loss weights than they do with L_ϵ?

Hint: Express L_x and L_s as multiples of L_ϵ. Compare low noise ($b_t \approx 0$) with high noise ($a_t \approx 0$).

(c) Discuss when each output is useful:

- **Noise:** What is convenient about the distribution of its training target? Compare its variance with that of the score target $-\boldsymbol{\epsilon}/b_t$.
- **Clean image:** Why is this output convenient for viewing predictions or comparing predicted and true pixel values? Use Problem 1(c) to give a limitation.
- **Score:** Why is this output convenient for the sampling formula in Problem 2(c)? Does this imply that training a score network gives better samples than training a noise network?

Reading. For the three prediction forms, see Luo (2022), *Understanding Diffusion Models: A Unified Perspective*, Section 3.2. For training losses and prediction choices, see Ho, Jain and Abbeel (2020), *DDPM*, Sections 3.4 and 4.2.

Optional exercises

These exercises are for further practice after the tutorial.

Optional A. Why does the average conditional score equal the marginal score? Let $p_{\text{data}}(\mathbf{x}_0)$ be the density of clean images. Starting from

$$q_t(\mathbf{z}_t) = \int p_{\text{data}}(\mathbf{x}_0) q(\mathbf{z}_t \mid \mathbf{x}_0)\, d\mathbf{x}_0,$$

show that

$$\nabla_{\mathbf{z}_t} \log q_t(\mathbf{z}_t) = \mathbb{E}[\nabla_{\mathbf{z}_t} \log q(\mathbf{z}_t \mid \mathbf{x}_0) \mid \mathbf{z}_t, t].$$

Steps: Write $\nabla_{\mathbf{z}_t} \log q_t(\mathbf{z}_t) = \nabla_{\mathbf{z}_t} q_t(\mathbf{z}_t)/q_t(\mathbf{z}_t)$. Move the gradient inside the integral (assume this is valid). Replace $\nabla_{\mathbf{z}_t} q(\mathbf{z}_t \mid \mathbf{x}_0)$ with $q(\mathbf{z}_t \mid \mathbf{x}_0)\nabla_{\mathbf{z}_t} \log q(\mathbf{z}_t \mid \mathbf{x}_0)$. Use Bayes' rule to find $q(\mathbf{x}_0 \mid \mathbf{z}_t, t)$ in the integral, then write the result as a conditional expectation.

Optional B. Clean-image prediction and the Gaussian KL. For $t \geq 2$, Lecture 11 gave the target reverse mean

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{z}_t, \mathbf{x}_0) = c_t \mathbf{x}_0 + d_t \mathbf{z}_t,$$

with coefficients

$$c_t = \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t},$$
$$d_t = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}.$$

The KL from the target reverse Gaussian to the model reverse Gaussian is

$$D_{\text{KL}}[q(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0) \,\|\, p_\theta(\mathbf{z}_{t-1} \mid \mathbf{z}_t)]$$
$$= \frac{1}{2\sigma_t^2} \|\tilde{\boldsymbol{\mu}}_t(\mathbf{z}_t, \mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{z}_t, t)\|^2 + C_t.$$

The reverse variance $\sigma_t^2 \mathbf{I}$ is fixed; C_t does not depend on θ. The forward variance b_t^2 and reverse variance σ_t^2 are different quantities.

(a) Replace $\mathbf{x}_0$ by $\hat{\mathbf{x}}_\theta(\mathbf{z}_t, t)$ to define $\boldsymbol{\mu}_\theta(\mathbf{z}_t, t) = c_t \hat{\mathbf{x}}_\theta(\mathbf{z}_t, t) + d_t \mathbf{z}_t$. Show that this equals the noise-prediction mean in Problem 2(c).

Hint: Substitute Problem 1(a), collect the $\mathbf{z}_t$ and $\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)$ coefficients, and use $\bar{\alpha}_t = \alpha_t \bar{\alpha}_{t-1}$.

(b) Show that the same Gaussian KL can be written as

$$D_{\text{KL}} - C_t = \frac{c_t^2}{2\sigma_t^2} L_x = w_t L_\epsilon = w_t b_t^2 L_s,$$

where

$$w_t = \frac{(1-\alpha_t)^2}{2\sigma_t^2 \alpha_t b_t^2}.$$

Steps: Subtract the target and model means. Substitute the result into the KL, then use Problem 3(a) to rewrite the loss and simplify the coefficients.